Noise Covariance as Representation Geometry: One Metric, Four Architecture Classes

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Abstract

Label-free scientific reconstruction learns representations from measured observations, without class labels or clean targets, and instrument-characterized noise covariance supplies the prior for deciding which residuals are informative. For Gaussian noise, inverse covariance is the likelihood-consistent reconstruction metric. Every ingredient here is classical — generalized least squares, the matched filter, weighted PCA, the linear-autoencoder/PCA correspondence, multilinear PCA — and we introduce no new estimator; we organize them as a pair $(\Sigma^{-1}, \mathcal{F})$, the likelihood fixing the residual geometry and the architecture the admissible representation class. Holding the metric fixed, we walk $\mathcal{F}$ up four rungs: optimal filtering, an unrestricted rank-k linear projection, a Kronecker-separable two-factor projection, and factored attention. At the linear rungs the metric question has a closed-form answer and reverses in practice: two models sharing an architecture and differing only in the training metric rank oppositely under raw MSE and the likelihood residual, across ten seeds with paired intervals excluding zero and a twenty-seed white-noise control at exactly zero. At the attention rung the reversal does not appear, because the two objectives stop separating once whitening approaches a constant rescaling. The metric fixes the geometry; whether that choice bites is a property of the instrument and the representation.

1 Introduction

Many scientific instruments produce abundant measurements but no class labels or clean targets, so each observation is used as its own training signal while an instrument-derived noise model says which discrepancies are meaningful. Such observations are continuous, multi-channel time series whose uncertainty varies across frequencies, sensors, and acquisition conditions. Standard mean-squared error (MSE) treats all residual coordinates equally, implicitly assuming independent noise of equal variance; under colored or correlated noise a large residual can lie in a poorly measured direction while a small one in a low-noise direction is statistically decisive. The measurement covariance is therefore a measurement-model prior fixing the geometry of reconstruction error, distinct from a conservation law, equivariance, or simulator constraint [Bronstein et al., 2021, Karniadakis et al., 2021].

This complements self-supervised reconstruction that learns without clean targets [Batson and Royer, 2019, Yaman et al., 2020, Gruber et al., 2025], and differs from the dominant self-supervised paradigm [Chen et al., 2020, He et al., 2022], which defines its signal through augmentation invariance or masked prediction and so presupposes nuisance structure that can be simulated; measurement noise supplies that structure directly.

Every individual link here is classical, and we state the boundary plainly. Inverse-covariance weighting is generalized least squares [Aitken, 1936] and, at rank one, the matched or optimal filter [Turin,

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1960, Kay, 1993, Szymkowiak et al., 1993]; that literature fixes the weighting but assumes a *known*, one-dimensional template. The latent-variable line relaxes the signal model but not the noise model: probabilistic PCA assumes isotropic noise [Tipping and Bishop, 1999] and factor analysis a diagonal covariance [Rubin and Thayer, 1982, Roweis, 1998], while weighted low-rank models [Bailey, 2012, Allen et al., 2014] and the linear-autoencoder/PCA correspondence [Baldi and Hornik, 1989] are established, as is multilinear PCA [Lu et al., 2008]. *We introduce no new estimator.* What is new is the arrangement: placing these in one Σ^{-1} metric as a ladder of admissible representation classes, the exact condition under which the Kronecker restriction is lossless (Proposition 2), and the observation that a factored-attention network is that same restriction relaxed to a nonlinear map — so the metric question does not stop where closed-form analysis does.

We ask: *what geometry should a label-free scientific reconstruction use under structured noise, and how should that geometry interact with the architecture that expresses the representation?* We organize reconstruction methods as a pair $(\Sigma^{-1}, \mathcal{F})$: the measurement likelihood fixes the residual geometry Σ^{-1}, while the architecture fixes the admissible representation class $\mathcal{F}$. Holding Σ^{-1} fixed, we walk $\mathcal{F}$ up a ladder of four architecture classes: a rank-one fixed template (optimal filtering, Section 2); an unrestricted rank-k linear projection (CW-PCA and its tied noise-aware autoencoder, Proposition 1); a Kronecker-separable two-factor projection (NFPA, Proposition 2); and factored attention. The first three rungs admit closed-form analysis; the fourth is settled empirically. The metric question recurs across the closed-form rungs: under non-isotropic noise, raw MSE and the likelihood residual can rank equal-capacity models in opposite orders (Section 2).

A reconstruction method is therefore summarized by the pair

$$\big(\Sigma^{-1},\ \mathcal{F}\big), \tag{1}$$

and the present paper establishes the linear covariance geometry (Section 2), the four-rung ladder (Section 3), and the evidence for each rung (Section 4); nonlinear representations, distribution shift, and adaptive sampling remain future work.

2 Covariance-weighted representation geometry

Let $x_i = s_i + n_i \in \mathbb{R}^d$ be a mean-subtracted observation with $n_i \sim \mathcal{N}(0, \Sigma)$ and $\Sigma \succ 0$. The training mean is estimated once, applied to held-out observations, and added back after reconstruction. For a candidate reconstruction $\hat{s}_i$, the Gaussian negative log-likelihood is, up to constants,

$$\chi^2(\hat{s}_i) = (x_i - \hat{s}_i)^\top \Sigma^{-1} (x_i - \hat{s}_i). \tag{2}$$

Whitening, $\tilde{x}_i = \Sigma^{-1/2} x_i$, maps this quadratic form to $\|\tilde{x}_i - \hat{\tilde{s}}_i\|_2^2$: the noise-aware problem is Euclidean projection in whitened coordinates and Σ^{-1}-orthogonal projection in measurement coordinates. We report the mean per-trace residual $N^{-1} \sum_i \chi_i^2$ or the per-coordinate normalized residual

$$\bar{\chi}^2 = \frac{1}{Nd} \sum_{i=1}^{N} \chi^2(\hat{s}_i), \tag{3}$$

stating explicitly which normalization is used.

The theory treats Σ as specified; in practice it comes from instrument calibration or noise-only records, is estimated and regularized without evaluation data, and a misspecified Σ defines a misspecified geometry (Appendix B.2).

Proposition 1 (Noise-weighted projection bridge)**.** *Under the model above, OF, rank-k covariance-weighted PCA (CW-PCA), and a tied noise-aware linear autoencoder minimize Eq.* (2) *over different admissible subspaces. OF fixes a rank-one template; CW-PCA learns a rank-k subspace; and the tied autoencoder parameterizes the same learned projection when its encoder is the Σ^{-1}-adjoint of its decoder.*

Proof sketch. Whitening reduces the problem to ordinary rank-k approximation, so every minimizer spans a leading eigenspace of the whitened sample covariance; at rank one this is the optimal filter, and the tied Σ^{-1}-adjoint encoder follows from the same geometry rather than from an architectural choice. Unlike the latent-variable precedents of Section 1, the full covariance is kept specified rather than assumed isotropic or diagonal. Details in Appendix A.

Metric reversal. Raw MSE replaces Σ^{-1} by I and is likelihood-correct only when $\Sigma = \sigma^2 I$. With colored or correlated noise, the two objectives rank residual directions differently: MSE follows raw variance, whereas Eq. (2) emphasizes high-information directions. Consequently, two equal-rank models can satisfy $\mathrm{MSE}(A) < \mathrm{MSE}(B)$ but $\chi^2(A) > \chi^2(B)$, with learned subspaces nearly orthogonal after whitening. This is not an evaluation paradox: each model is optimal in a different geometry.

3 The architecture ladder

Holding the metric fixed, we walk the representation class $\mathcal{F}$ of Eq. (1) up four rungs. Rungs 1–3 are analysed in general two-axis form and evidenced synthetically in Section 4; rung 4 is evidenced on the short-time Fourier transform (STFT) band–frame plane of a public benchmark, where Σ^{-1} is diagonal in frequency by measurement (Section 4).

Rung 1: rank-one fixed template (optimal filtering). Fixing the signal to a known template s_0, the likelihood-optimal reconstruction is the optimal filter (OF) amplitude of Section 2 [Gatti and Manfredi, 1986, Turin, 1960]. The representation class is a single direction, and the metric question has a closed-form answer.

Rung 2: unrestricted rank-k linear. Replacing the fixed template with a learned rank-k subspace gives covariance-weighted PCA (CW-PCA), which is also the tied noise-aware linear autoencoder: both minimize the likelihood residual over the full Grassmannian $\mathrm{Gr}(k, n_1 n_2)$ of a two-axis measurement (Proposition 1), with reconstruction $\hat{s} = PP^\top \Sigma^{-1} x$.

Rung 3: Kronecker-separable two-axis projection (NFPA). NFPA restricts rung 2's projection to separable factors of a two-axis measurement $Z_i \in \mathbb{R}^{n_1 \times n_2}$ — sensor channel $\times$ time in the controlled synthetic study, frequency band $\times$ time frame on the public benchmark. It learns orthonormal bases $U_1 \in \mathbb{R}^{n_1 \times k_1}$ and $U_2 \in \mathbb{R}^{n_2 \times k_2}$, with code $C_i = U_1^\top Z_i U_2$ and reconstruction $\hat{Z}_i = U_1 U_1^\top Z_i U_2 U_2^\top$, so that

$$\mathrm{vec}(\hat{Z}_i) = \big(U_2 U_2^\top \otimes U_1 U_1^\top\big)\mathrm{vec}(Z_i), \tag{4}$$

and the joint representation is $\mathrm{range}(U_1) \otimes \mathrm{range}(U_2)$. The admissible family is the embedded product

$$\mathrm{Gr}(k_1, n_1) \times \mathrm{Gr}(k_2, n_2) \hookrightarrow \mathrm{Gr}(k_1 k_2, n_1 n_2). \tag{5}$$

Proposition 2 (Kronecker-restricted bridge)**.** *NFPA minimizes the CW-PCA/tied-autoencoder objective restricted to Kronecker-separable two-axis subspaces. Its optimum equals the unrestricted CW-PCA optimum if and only if at least one rank-$k_1 k_2$ CW-PCA minimizer has such a separable subspace.*

Proof sketch. Equation (4) is the orthogonal projector onto $\mathrm{range}(U_1) \otimes \mathrm{range}(U_2)$ in whitened coordinates, so NFPA is the whitened rank-$k_1 k_2$ objective restricted to Eq. (5); equality holds precisely when that set contains an unrestricted minimizer, and otherwise the gap is the cost of the architecture prior, not a failure of the metric. This is a covariance-aware member of the multilinear-PCA family [Lu et al., 2008], not a new factorization. When either factor is full rank the restriction is inactive and rung 3 reduces to rung 2; at $k_1 = k_2 = 1$ it recovers rung 1 (Corollary 1, Appendix A).

Rung 4: factored attention. Rung 3's two-factor restriction, instantiated on the band–frame plane, reads as $\mathrm{range}(U_f) \otimes \mathrm{range}(U_m)$ — separable frequency-band and time-frame factors; rung 4's transformer performs intra-band temporal attention over frames, then cross-band attention. These are the same factorization at different levels of nonlinearity, in the same coordinates, and Σ^{-1} is diagonal in exactly the basis the first factor indexes: on this benchmark a stationary-PSD model fits the readout noise ($\chi^2/\mathrm{dof} = 1.0002$, all gates passed; Section 4). The rung-4 model is a transformer [Vaswani et al., 2017] over frame patches [Nie et al., 2023], with rotary embeddings on the temporal axis [Su et al., 2024] and learned embeddings across bands; the metric question is posed empirically, training identical architecture, data, and budget on two objectives differing only in the metric (Section 4). Injecting the optimal-filter statistic into the forward pass follows prior work on calorimetric event reconstruction [Sjostrom, 2026] and serves as a secondary contrast.

4 Evidence

Metric reversal in the linear rungs. A 30-seed rank-two synthetic benchmark and a ten-seed white-noise control establish the reversal at rung 2, and on one public TIDMAD file with an injected reference [Fry et al., 2024] the weighted residual and the recorded truth agree against raw MSE — one file at one operating point, not replicated at a confirmatory file. Numbers in Appendix B.2. The consolidated evidence below supersedes these as the paper's linear-rung result.

Why the architecture binds next. TIDMAD's diagnostics show the real signal is not a fixed low-dimensional subspace — the tone sweeps, one file is two-tone, and the analytic rank-two prediction failed on three files for diagnosed non-implementation reasons — so the architecture, not the metric, is what binds; rung 4 lifts it.

NFPA under channel compression. Rung 3 is consolidated on a non-degenerate configuration with $C = 8 > k_c = 2$ channels and matched total rank $k = k_c k_t = 6$: NFPA, an isotropically trained factored baseline of the same architecture, CW-PCA, and ordinary PCA are fitted at matched rank on the same ten seeds. Because NFPA and the isotropic baseline share an architecture and differ only in the training metric, their paired contrast isolates the metric. It reverses: NFPA is better under the likelihood residual (paired difference -0.0192, seed-level bootstrap 95% interval $[-0.0194, -0.0190]$) and worse under raw MSE ($+0.0034$, $[+0.0034, +0.0034]$), both intervals excluding zero over ten seeds. Against the unrestricted CW-PCA the held-out likelihood residual differs by -0.0026 ($[-0.0026, -0.0025]$); this is a finite-sample held-out difference, not evidence that the Kronecker restriction is intrinsically superior. In the twenty-seed white-noise control both contrasts are zero to machine precision (0.000000, interval $[0, 0]$), isolating covariance geometry rather than rank or optimization as the cause of the separation. The single-split $C = k_c = 2$ run is retained in Appendix B.2.

Rung 4: the metric in a factored-attention network. One factored-attention architecture is trained on the TIDMAD readout with two objectives differing only in the metric — `T-mse` on the unwhitened spectrogram, `T-chi2` on the Σ^{-1}-weighted residual — at matched data, budget, seed handling and evaluation grid. *The reversal does not appear at this rung.* Over fifty held-out evaluations the Σ^{-1}-trained arm is better on *both* criteria: paired difference -1.15×10^{-4} on the normalized likelihood residual ($[-1.27, -1.03] \times 10^{-4}$) and -57.2 on raw MSE ($[-71.6, -42.6]$), both intervals excluding zero but the second in the direction opposite to prediction. The release's coarse denoising score does not separate the arms: the paired difference is -1.0×10^{-3} with window and file-block bootstrap intervals of $[-3.3, +1.4]$ and $[-4.6, +2.4] \times 10^{-3}$, both spanning zero, and both arms sit far below the published baselines (Appendix B.1); Figure 1 shows all three contrasts together. The likely reason the objectives fail to separate is that whitening the band-limited spectrogram is close to a constant rescaling over the modelled band, leaving the two losses nearly collinear — a property of this instantiation rather than of the ladder.

Covariance adequacy on real data. We check calibration on a public gravitational-wave strain release [LIGO Scientific Collaboration and Virgo Collaboration, 2016, Macleod et al., 2021]. The whitened score matches its reference form to 10^{-12} or better and is calibrated on stationary synthetic noise (dispersion 1.15), yet on real H1 and L1 strain it is seven to nine times over-dispersed (9.42, 8.83), which a global leave-one-out variant does not rescue (1.77, 7.29). A covariance model can be correctly implemented, validated synthetically, and still fail calibration on real data: implementation identity is not calibration.

Limitations. The ladder asks whether the likelihood geometry matters and how it interacts with the architecture, answered in closed form at the linear rungs and empirically at the factored-attention rung, where it is not observed. Four caveats bound that evidence. Rung 4 is one seed per arm, so its held-out differences are contrasts over evaluations rather than evidence about seed variability; both arms sit far below the published baselines; and the near-collinearity account is inferred from the effect signature, not measured. The linear-rung TIDMAD reversal is not replicated at a confirmatory file. Stationary-PSD adequacy is per-instrument: adequate on the TIDMAD readout ($\chi^2/\mathrm{dof} = 1.0002$), failing on gravitational-wave strain. The $C = 8 > k_c = 2$ result is synthetic with an oracle covariance, bounding what a correctly specified Σ buys rather than an estimated one.

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A Proof details

A.1 Noise-weighted projection bridge

Let $W = \Sigma^{-1/2}P$. The constraint $P^\top\Sigma^{-1}P = I_k$ becomes $W^\top W = I_k$, and the reconstruction satisfies

$$\Sigma^{-1/2}\hat{s} = \Sigma^{-1/2}PP^\top\Sigma^{-1}x = WW^\top\tilde{x}.$$

The empirical loss is therefore $N^{-1}\sum_i \|\tilde{x}_i - WW^\top\tilde{x}_i\|_2^2$. By Rayleigh–Ritz, every minimizer spans a leading rank-k eigenspace of the whitened sample covariance. Mapping back yields the CW-PCA subspace. The objective depends on W only through $WW^\top$, so W is identified up to an orthogonal rotation.

A.2 Kronecker-restricted bridge

For every $Z \in \mathbb{R}^{C\times T}$, the vectorization identity gives

$$\mathrm{vec}(U_cU_c^\top ZU_tU_t^\top) = (U_tU_t^\top \otimes U_cU_c^\top)\mathrm{vec}(Z).$$

The Kronecker product is a symmetric idempotent matrix of rank k_ck_t, hence the orthogonal projector onto $\mathrm{range}(U_t)\otimes\mathrm{range}(U_c)$. NFPA is thus the ordinary whitened rank-k_ck_t projection loss restricted to product subspaces. The restricted and unrestricted minima coincide exactly when an unrestricted minimizing projector belongs to this subset.

A.3 Rank-one correspondence

For $Z_i = A_ivt^\top + \epsilon_i$ with isotropic whitened noise and a unique leading signal eigendirection separated by a nonzero eigengap, the population second moment has leading vectorized direction $(t\otimes v)/(\|t\|\|v\|)$, which is feasible for rank-one NFPA. Its projection coefficient is proportional to $\langle \mathrm{vec}(Z_i), t\otimes v\rangle$, the whitened matched-filter coefficient and hence the OF amplitude up to normalization.

Corollary 1 (Rank-one correspondence)**.** *Assume isotropic whitened noise and a unique leading population signal eigendirection separated by a nonzero eigengap. If that direction is generated by a separable template $\tilde{s} = vt^\top$, population NFPA with $k_1 = k_2 = 1$ learns $v/\|v\|$ and $t/\|t\|$ up to signs; its scalar code is proportional to the OF amplitude of Section 2.*

B Experiment details and reporting limits

The controlled NFPA experiment uses $C = 2$ channels, $T = 256$ samples, a sampling rate of $1\,\mathrm{kHz}$, channel weights $(1.0, 0.65)$, and amplitude $A \sim \mathcal{N}(1, 0.3^2)$. The noise is Brownian with $S_n(f) \propto 1/f^2$ and is whitened using the matched temporal frequency-domain filter. The train and test sets contain 800 and 300 events. NFPA and IsoAE use $(k_c, k_t) = (2, 3)$, while CW-PCA and IsoPCA use rank six.

Because $C = k_c = 2$, $U_cU_c^\top = I_2$ in this experiment: the reconstruction restricts a temporal subspace shared by both channels but does not compress the channel axis. Principal angles between NFPA and CW-PCA are $0.0°$, $59.3°$, $64.6°$, $73.8°$, $84.8°$, and $87.8°$ (mean $61.7°$). The corresponding IsoAE–CW-PCA angles are $82.1°$, $83.7°$, $86.0°$, $86.6°$, $88.2°$, and $89.4°$ (mean $86.0°$). These are point estimates from one seed and split. The exact held-out raw-MSE values for IsoAE and IsoPCA are re-exported at full precision from the archived run (4.58×10^{-9} and 4.33×10^{-9}).

B.1 TIDMAD rung-4 baselines

A submission-strength empirical study should repeat this comparison across seeds and train/test splits and sweep SNR, noise color, channel count, rank pair, and controlled nonseparability. It should report confidence intervals for held-out $\bar{\chi}^2$, raw MSE, and subspace angles. No such uncertainty estimate is claimed here.

Table 1: Implementation check: the single-split $C = k_c = 2$ run reported by the earlier draft, retained for provenance. $\bar{\chi}^2$ is the per-coordinate normalized whitened residual from Eq. (3); $\bar{\theta}$ is the mean principal angle to CW-PCA in whitened space.

Method	Training	Class	$\bar{\chi}^2$	Raw MSE	$\bar{\theta}$
CW-PCA	Σ^{-1}	full	0.9929	4.95×10^{-3}	0°
NFPA	Σ^{-1}	factored	0.9916	4.95×10^{-3}	61.7°
IsoAE	MSE	factored	0.9995	4.58×10^{-9}	86.0°
IsoPCA	MSE	full	1.0007	4.33×10^{-9}	—

Table 2: TIDMAD coarse denoising scores (higher is better) from published baselines, the archived linear estimate, and this paper's two rung-4 arms, which share architecture, data, and budget and differ only in the training objective.

Method	Coarse score
FCNet	6.55
Transformer	4.18
PUNet	3.84
none	1.10 (1.1026)
linear Wiener (archived)	-1.8025
`T-mse`	**[PENDING: coarse score — PLAN_03]**
`T-chi2`	**[PENDING: coarse score — PLAN_03]**

B.2 Predeclared confirmatory protocol

The multi-seed confirmatory study is predeclared before execution and tests three hypotheses. H1 (metric reversal at fixed architecture): under colored or correlated noise, NFPA and an isotropically trained multilinear projection with the same (k_c, k_t) should disagree in ranking, with paired across-seed confidence intervals excluding zero in the predicted directions. H2 (white-noise control): when $\Sigma = \sigma^2 I$, covariance-aware and Euclidean projectors should agree to numerical tolerance. H3 (active restriction): with $C > k_c$ and $T > k_t$, NFPA should recover the planted channel and temporal subspaces for separable signals. Truth-referenced recovery (an earlier fourth hypothesis) is now covered by the rung-4 amplitude error against TIDMAD's real injected reference channel and is not run synthetically. The primary configuration uses $C = 8$ channels, $T = 256$ samples, ranks $(k_c, k_t) = (2, 3)$, $800/400/2000$ training/validation/test events, 10 independent seeds, 10 ALS restarts per seed, SNR 10 under $\mathrm{SNR}^2 = \mathbb{E}[s^\top \Sigma^{-1} s]$, Brownian $1/f^2$ temporal noise with channel correlation, oracle covariance, and nonseparability $\alpha = 0.1$. Results enter the main text only if every acceptance gate holds: all seeds complete without post-result configuration changes; the fixed-architecture reversal intervals exclude zero in the predicted directions; the white-noise control passes its predeclared equivalence margin; and the table and figure are generated directly from archived seed-level results.

The factored representation (Section 3, generic form) has $O(k_1 n_1 + k_2 n_2)$ basis parameters rather than $O(k_1 k_2 n_1 n_2)$ for an unstructured basis, trading approximation flexibility for a two-axis inductive bias related to classical tensor factorizations [Kolda and Bader, 2009, Lu et al., 2008]. Alternating spectral updates solve each factor subproblem exactly, although the joint product-manifold objective remains nonconvex.

B.3 Evidence figure

Rung-2 numbers carried from the companion work. In the 30-seed rank-two benchmark (512 traces of dimension 256 per seed), PCA and CW-PCA had held-out weighted residuals 48.419 and 47.869 (paired difference 0.550, $[0.530, 0.571]$) while raw MSE preferred the opposite reconstruction, their whitened subspaces 86.25° apart. The ten-seed white-noise control gave a maximum angle of 1.48×10^{-6} degrees. On the TIDMAD file the amplitude RMSE fell from 0.1209 to 0.0880.

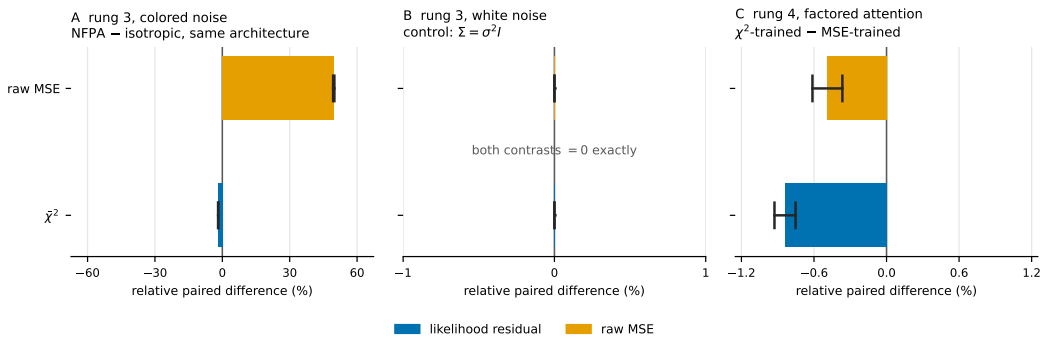


Figure 1: Paired contrasts at fixed architecture, as relative differences so the rungs are comparable. **A**, rung 3 under colored noise, ten seeds: NFPA minus the isotropically trained factored baseline is -1.86% on the normalized likelihood residual and $+49.6\%$ on raw MSE. The two point in opposite directions — metric reversal, with both intervals excluding zero. **B**, the same contrast with $\Sigma = \sigma^2 I$ over twenty seeds is exactly zero, isolating covariance geometry rather than rank or optimization as the cause of A. **C**, rung 4, fifty held-out evaluations at a matched step grid: the Σ^{-1}-trained arm is better on *both* criteria (-0.84% and -0.49%), so the reversal does not survive into this instantiation. Intervals are seed-level (A, B) or evaluation-level (C) percentile bootstraps, 10^4 resamples. Every value is computed from the archived per-seed and per-evaluation records by `scripts/make_rps_figure.py`.